

Integrated Homeostatic Systems

Fluid, acid-base, temperature, metabolism, oxygen delivery, autonomic control, sleep, stress, exercise, shock, and reserve

Volume 03O of the Health Atlas Research Suite

Purpose

Provide the cross-organ physiology and measurement architecture required to understand stability, adaptation, challenge, decompensation and recovery.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not provide resuscitation, electrolyte correction, ventilator, glucose-crisis or temperature-treatment protocols. These states can deteriorate rapidly and require condition-specific professional care.

How to use this volume

Homeostasis is not static normality. It is coordinated control across organs, behavior and environment, with reserve revealed under challenge and recovery. The same “normal” resting value can coexist with fragile compensation, while an abnormal value can be adaptive in context. A complete profile therefore combines coupled variables, timing, perturbation, reserve, symptoms, organ function and treatment effects.

Questions this volume is designed to answer

1. How do organ systems coordinate stability and adaptation?
2. How should reserve and recovery be measured under challenge?
3. Why can one biomarker or composite score misrepresent whole-system state?
4. How should multimodal monitoring and closed-loop control be validated?
5. What separates useful early warning from alarm burden and false reassurance?

Scope

- Feedback/control, fluid/osmolality, acid-base, thermoregulation, energy, oxygen delivery, autonomic, sleep/circadian and stress physiology.
- Shock, sepsis, thermal illness, metabolic/electrolyte crises, dysautonomia, fatigue, frailty and multiorgan failure.
- Vitals, gases/labs, hemodynamics, CPET, autonomic/sleep, wearables and closed-loop systems.
- Acute restoration, environmental prevention, rehabilitation and recovery systems.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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Executive synthesis

Homeostasis is not static normality. It is coordinated control across organs, behavior and environment, with reserve revealed under challenge and recovery. The same “normal” resting value can coexist with fragile compensation, while an abnormal value can be adaptive in context. A complete profile therefore combines coupled variables, timing, perturbation, reserve, symptoms, organ function and treatment effects.

Central organizing principle

Provide the cross-organ physiology and measurement architecture required to understand stability, adaptation, challenge, decompensation and recovery.

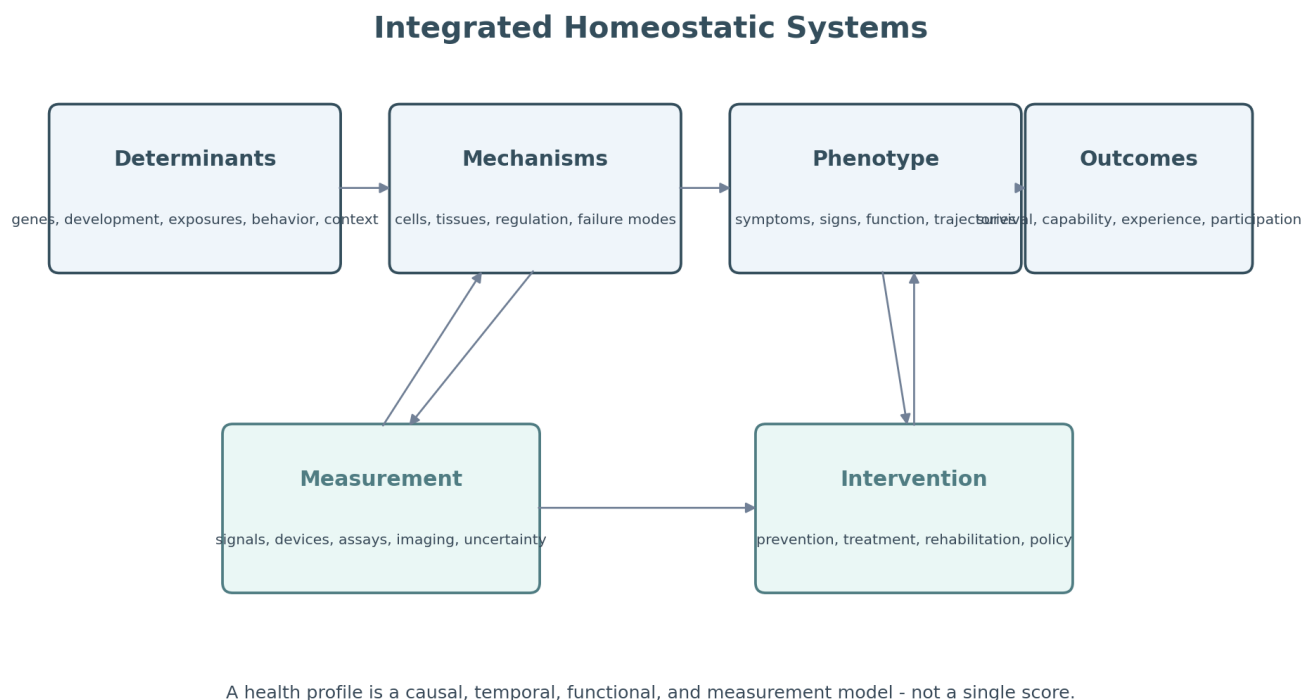


Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Altered consciousness, severe breathing difficulty, chest pain, shock signs, extreme temperature, seizure, severe glucose/electrolyte disturbance or rapid multiorgan deterioration are emergencies. Closed-loop human systems require fault-tolerant safety engineering and clinical oversight.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early knowledge and practice	Early medicine recognized balance, fever, pulse, thirst, sleep and crisis as whole-body phenomena.	Preserved integrated observation.	Humoral explanations lacked measurable mechanisms.
Anatomical and demographic era	Circulation, respiration, metabolism, thermodynamics and internal-environment concepts established quantitative physiology.	Made regulation and organ coupling testable.	Laboratory averages often displaced individual dynamics.
Hormonal, laboratory and surgical era	Blood gases/electrolytes, intensive care, dialysis, ventilation and hemodynamic monitoring enabled active support.	Transformed survival in reversible failure.	Device targets sometimes encouraged treatment of numbers rather than causes.
Rights, public-health and evidence era	Control theory, exercise physiology, sleep science, sepsis definitions and rehabilitation expanded challenge/recovery models.	Connected baseline to reserve and function.	Specialties still partition coupled failures.
Molecular, assisted and digital era	Wearables, continuous sensors, computational physiology and closed-loop devices increase temporal resolution.	May detect transitions and personalize support.	Artifacts, opaque models and alarm/automation harms are substantial.
Current systems model	Current homeostatic models integrate physiology, behavior, environment, allostasis, resilience and person-valued recovery.	Provides a bridge across all organ volumes.	No complete digital twin or universal health score has clinical validity.

2.0.1 Early medicine recognized balance, fever, pulse, thirst, sleep and crisis as whole-body phenomena.

Scientific consequence. Preserved integrated observation.

What remained unresolved. Humoral explanations lacked measurable mechanisms.

Evidence anchors: [3,5]

2.0.2 Circulation, respiration, metabolism, thermodynamics and internal-environment concepts established quantitative physiology.

Scientific consequence. Made regulation and organ coupling testable.

What remained unresolved. Laboratory averages often displaced individual dynamics.

Evidence anchors: [3,5]

2.0.3 Blood gases/electrolytes, intensive care, dialysis, ventilation and hemodynamic monitoring enabled active support.

Scientific consequence. Transformed survival in reversible failure.

What remained unresolved. Device targets sometimes encouraged treatment of numbers rather than causes.

Evidence anchors: [3,5]

2.0.4 Control theory, exercise physiology, sleep science, sepsis definitions and rehabilitation expanded challenge/recovery models.

Scientific consequence. Connected baseline to reserve and function.

What remained unresolved. Specialties still partition coupled failures.

Evidence anchors: [3,5]

2.0.5 Wearables, continuous sensors, computational physiology and closed-loop devices increase temporal resolution.

Scientific consequence. May detect transitions and personalize support.

What remained unresolved. Artifacts, opaque models and alarm/automation harms are substantial.

Evidence anchors: [3,5]

2.0.6 Current homeostatic models integrate physiology, behavior, environment, allostasis, resilience and person-valued recovery.

Scientific consequence. Provides a bridge across all organ volumes.

What remained unresolved. No complete digital twin or universal health score has clinical validity.

Evidence anchors: [1,5]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Feedback, feedforward and redundancy

Core principle. Physiologic variables are kept within viable, state-dependent ranges through multiple sensors, controllers and effectors.

Control architecture. Negative/positive feedback, prediction, hierarchical control, reserve, redundancy and adaptation.

Failure modes. Oscillation, runaway feedback, delayed response, brittle compensation or hidden reserve loss.

Observable consequences. Baseline, variability, response to challenge, recovery, coupled variables and outcomes.

Cross-system coupling. All organ systems and behavior/environment.

Evidence anchors: [3,5,6]

3.0.2 Fluid volume and osmolality

Core principle. Water and electrolytes distribute across compartments while kidneys, thirst and hormones regulate effective circulation and cell volume.

Control architecture. ADH, renin-angiotensin-aldosterone, natriuretic signals, thirst, kidney and vascular tone.

Failure modes. Dehydration, overload, hypo/hypernatremia, edema, shock or cellular swelling.

Observable consequences. Intake/output, weight, vitals, electrolytes/osmolality, urine, exam and hemodynamics.

Cross-system coupling. Renal, cardiovascular, endocrine and neurologic systems.

Evidence anchors: [2,3,10]

3.0.3 Acid-base regulation

Core principle. Buffers, lungs and kidneys control hydrogen ion while metabolism generates volatile and fixed acids.

Control architecture. Bicarbonate/CO₂, ventilation, renal reabsorption/generation, proteins/phosphate and tissue perfusion.

Failure modes. Metabolic/respiratory acidosis or alkalosis and mixed disorders.

Observable consequences. Blood gas, electrolytes, anion gap, lactate/ketones, ventilation and clinical cause.

Cross-system coupling. Respiratory, renal, metabolic and critical-care systems.

Evidence anchors: [3,4,10]

3.0.4 Thermoregulation

Core principle. Core temperature reflects heat production, distribution and loss controlled by neural and behavioral responses.

Control architecture. Hypothalamus, skin blood flow, sweating, shivering, metabolism, clothing and environment.

Failure modes. Heat exhaustion/stroke, hypothermia, fever dysregulation or medication-related syndromes.

Observable consequences. Core/peripheral temperature, environment, hydration, cognition, organs and recovery.

Cross-system coupling. Skin, cardiovascular, nervous, endocrine and occupational systems.

Evidence anchors: [3,16,17]

3.0.5 Energy balance and substrate switching

Core principle. Cells and organs coordinate intake, storage, glucose/lipid use and expenditure across feeding, fasting and exercise.

Control architecture. Insulin/glucagon, autonomic/endocrine signals, liver, muscle, adipose, mitochondria and behavior.

Failure modes. Hypoglycemia, ketoacidosis, starvation, obesity, cachexia or exercise intolerance.

Observable consequences. Glucose/ketones, indirect calorimetry, body composition, intake/activity and organ function.

Cross-system coupling. Endocrine, GI, muscle, brain and social systems.

Evidence anchors: [3,12,13]

3.0.6 Oxygen delivery and extraction

Core principle. Lungs load oxygen, blood carries it, heart/vessels distribute it and mitochondria use it.

Control architecture. Ventilation/perfusion, hemoglobin, cardiac output, microcirculation and cellular metabolism.

Failure modes. Hypoxemia, anemia, low flow, diffusion/extraction failure, lactate and organ dysfunction.

Observable consequences. Oximetry/blood gases, hemoglobin, flow/hemodynamics, lactate and organ response.

Cross-system coupling. Respiratory, hematologic, cardiovascular and metabolic systems.

Evidence anchors: [2,7,9]

3.0.7 Autonomic regulation and orthostatic adaptation

Core principle. Sympathetic, parasympathetic and local reflexes coordinate pressure, heart rate, digestion, temperature and stress responses.

Control architecture. Baroreflex, chemoreflex, vagal/sympathetic output, volume and central state.

Failure modes. Syncope, orthostatic intolerance, dysautonomia, inappropriate tachycardia or impaired recovery.

Observable consequences. Supine/upright vitals, ECG, autonomic tests, symptoms and functional challenge.

Cross-system coupling. Cardiovascular, nervous, endocrine, GI and behavioral systems.

Evidence anchors: [2,6,15]

3.0.8 Sleep and circadian coordination

Core principle. Central/peripheral clocks and sleep states organize endocrine, autonomic, immune, metabolic and cognitive recovery.

Control architecture. Light, melatonin, sleep pressure, behavior, work timing and organ clocks.

Failure modes. Insomnia, circadian misalignment, sleep apnea consequences, metabolic/mental impairment and accidents.

Observable consequences. History/diary, actigraphy, polysomnography when indicated, light/activity and daytime function.

Cross-system coupling. Brain, respiratory, endocrine, immune and occupational systems.

Evidence anchors: [6,14,17]

3.0.9 Stress response and allostasis

Core principle. Acute stress mobilizes cardiovascular, endocrine, immune and behavioral resources; repeated activation can produce cumulative cost.

Control architecture. HPA axis, autonomic, immune, metabolic, appraisal and social safety.

Failure modes. Chronic allostatic load, anxiety/trauma syndromes, hypertension/metabolic disease or immune dysregulation.

Observable consequences. Exposure/context, symptoms, sleep, physiology, function and longitudinal outcomes; no single allostatic marker.

Cross-system coupling. Psychosocial, brain/behavior, cardiovascular and immune systems.

Evidence anchors: [6]

3.0.10 Exercise and reserve under challenge

Core principle. Integrated cardiopulmonary, muscular, neural and metabolic systems increase output and recover after demand.

Control architecture. Central command, autonomic output, ventilation, circulation, substrate and thermoregulation.

Failure modes. Exercise intolerance, exertional hypoxemia/ischemia, chronotropic/ventilatory limitation, fatigue or post-exertional worsening.

Observable consequences. CPET, field tests, symptoms, ECG/gases, recovery and daily activity.

Cross-system coupling. Cardiovascular, respiratory, muscle, nervous and rehabilitation systems.

Evidence anchors: [2,7,20]

3.0.11 Inflammation, coagulation and repair in acute stress

Core principle. Injury/infection activates defense, endothelial/coagulation, endocrine and metabolic programs that must resolve.

Control architecture. Pattern recognition, cytokines, complement, clotting, neuroendocrine stress and tissue repair.

Failure modes. Sepsis, DIC, immune paralysis, fibrosis or multiorgan failure.

Observable consequences. Source/pathogen, organ function, perfusion, coagulation, immune trajectory and recovery.

Cross-system coupling. Immune, hematologic, all organs and critical-care systems.

Evidence anchors: [9,18]

3.0.12 Frailty, resilience and recovery

Core principle. Reserve and network redundancy determine vulnerability to stress and ability to return to baseline or adapt.

Control architecture. Age, multimorbidity, muscle/cognition, nutrition, social support and healthcare exposures.

Failure modes. Delirium, deconditioning, prolonged recovery, disability, institutionalization or death.

Observable consequences. Frailty/function, cognition, mobility, nutrition, social context and post-event trajectory.

Cross-system coupling. Life course, rehabilitation, whole-person and delivery systems.

Evidence anchors: [20,21,22]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Mass balance	$Accumulation = Input - Output + Production - Consumption$	General accounting law for water, solutes, heat, gases and metabolites.	Compartments, measurement error and unobserved fluxes must be specified.
Henderson-Hasselbalch for bicarbonate	$pH = 6.1 + \log_{10} \left(\frac{[HCO_3^-]}{0.03 P_{CO_2}} \right)$	Relates major extracellular buffer to CO ₂ under defined units/equilibrium.	Mixed disorders, non-bicarbonate buffers and sampling conditions require full interpretation.
Fick principle	$\dot{V}_{O_2} = Q (C_{aO_2} - C_{vO_2})$	Links oxygen consumption to flow and arteriovenous content difference.	Measurement assumptions and regional heterogeneity matter.
Heat balance	$S = M - W - E - R - C - K$	Stored heat equals metabolism minus external work and evaporative, radiative, convective and conductive losses.	Each term depends on clothing, environment, geometry and physiology.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within integrated homeostatic systems.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within integrated homeostatic systems.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[13]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[6,14]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[6]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[34]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: ^[16,17]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[35]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the integrated regulatory failure, challenge-response phenotype, acute syndrome, or reserve trajectory, but cross-system processes and functional effects are retained.

5.0.1 Shock syndromes

Definition and phenotype. Failure to deliver/utilize adequate oxygen and substrate due to low volume, pump failure, obstruction or distributive vascular dysfunction.

Pathophysiology. Reduced effective flow, maldistribution and cellular stress progress to organ injury.

Determinants and modifiers. Bleeding/fluid loss, sepsis, cardiac disease, embolism, anaphylaxis and treatment delay.

Assessment. Clinical perfusion, vitals/trends, lactate, ECG/echo/ultrasound, labs and cause-specific imaging.

Intervention logic. Immediate cause-directed resuscitation, source/bleeding/obstruction control and organ support while avoiding fluid/pressor harm.

Known limits and open questions. No single pressure or lactate defines all shock; macrocirculation can normalize while microcirculation remains impaired.

Suite links. 03A cardiovascular; 04 cross-system; 12 critical care.

Evidence anchors: [2,18,35]

5.0.2 Sepsis and multiorgan dysfunction

Definition and phenotype. Life-threatening organ dysfunction caused by dysregulated response to infection.

Pathophysiology. Immune, endothelial, coagulation, metabolic, autonomic and mitochondrial processes interact across organs.

Determinants and modifiers. Pathogen/source, host reserve, age, comorbidity, AMR, access and treatment timing.

Assessment. Syndrome recognition, cultures/molecular tests, organ/perfusion measures and source imaging; repeated reassessment.

Intervention logic. Antimicrobials/source control and individualized organ/hemodynamic support.

Known limits and open questions. Biomarker-defined subgroups and targeted host therapies remain uncertain.

Suite links. 08 infection; 03H immune; 03G coagulation.

Evidence anchors: [18,19,35]

5.0.3 Dehydration and volume depletion

Definition and phenotype. Net fluid/salt loss reduces tissue hydration and/or effective circulation.

Pathophysiology. GI/renal/skin losses or inadequate intake exceed compensatory thirst, hormones and renal conservation.

Determinants and modifiers. Heat, illness, infancy/aging, disability, diuretics, food/water insecurity and cognition.

Assessment. History/intake/output, weight, orthostatic/vitals, electrolytes/renal/urine and cause.

Intervention logic. Oral/IV replacement with composition/rate matched to severity and losses; treat cause.

Known limits and open questions. Physical signs and BUN/creatinine are nonspecific; overcorrection causes harm.

Suite links. 03F renal; 03E GI; 09 heat/environment.

Evidence anchors: [3,10,16]

5.0.4 Fluid overload and edema

Definition and phenotype. Excess total or maldistributed extracellular fluid from cardiac, renal, hepatic, venous/lymphatic or iatrogenic causes.

Pathophysiology. Sodium/water retention and hydrostatic/oncotic/permeability/lymphatic forces shift fluid into tissues.

Determinants and modifiers. Heart/kidney/liver disease, pregnancy, inflammation, medications and resuscitation.

Assessment. Weight, exam, ultrasound/hemodynamics, kidney/liver/cardiac tests and trajectory.

Intervention logic. Treat cause; sodium/fluid strategy and diuresis/renal replacement/compression where appropriate.

Known limits and open questions. Edema volume and intravascular filling can diverge.

Suite links. 03A, 03F, 03E liver, 03H lymphatic.

Evidence anchors: [2,11]

5.0.5 Sodium and osmolality disorders

Definition and phenotype. Hypo- or hypernatremia reflecting water balance relative to exchangeable solute, with neurologic risk.

Pathophysiology. ADH/thirst/kidney and intake/loss disorders change extracellular tonicity and brain cell volume.

Determinants and modifiers. Drugs, endocrine, renal/heart/liver disease, polydipsia, impaired access to water and iatrogenic fluids.

Assessment. Measured serum/urine osmolality, urine sodium, volume/cause context and symptom/chronicity.

Intervention logic. Cause-specific carefully rate-limited correction and monitoring.

Known limits and open questions. Volume assessment is imperfect and rapid correction can cause catastrophic neurologic injury.

Suite links. 03F renal; 03D endocrine; 03C nervous.

Evidence anchors: [3,10]

5.0.6 Acid-base disorders

Definition and phenotype. Primary metabolic or respiratory changes in pH/CO₂/bicarbonate with compensatory responses and possible mixed states.

Pathophysiology. Ventilation, renal handling, GI loss, metabolism, toxins and perfusion alter acid production/excretion.

Determinants and modifiers. Lung/kidney disease, sepsis, diabetes, vomiting/diarrhea, drugs and toxins.

Assessment. Blood gas and electrolytes with expected compensation, anion/osmolar gaps and cause-specific tests.

Intervention logic. Treat life-threatening physiology and cause; ventilation/fluids/electrolytes/antidotes/dialysis when indicated.

Known limits and open questions. Sampling and simplified rules can miss mixed or temperature-dependent states.

Suite links. 03B respiratory; 03F renal; 03D metabolic.

Evidence anchors: [3,4,10]

5.0.7 Heat illness

Definition and phenotype. Spectrum from heat edema/cramps/syncope/exhaustion to heat stroke with CNS and organ injury.

Pathophysiology. Environmental/metabolic heat exceeds dissipation; dehydration and circulatory strain worsen failure.

Determinants and modifiers. Heat/humidity, exertion/PPE, age, drugs, illness, acclimatization and inequitable work/housing.

Assessment. Core temperature using appropriate method, CNS/organ assessment, environment and exertional context.

Intervention logic. Immediate cooling for heat stroke and supportive organ care; prevention through exposure/work systems.

Known limits and open questions. Peripheral thermometers can falsely reassure and definitions vary.

Suite links. 09 environment/occupation; 03J skin; 03F kidney.

Evidence anchors: [16,17,35]

5.0.8 Hypothermia and cold injury

Definition and phenotype. Core cooling and local tissue freezing/nonfreezing injury impair neural, cardiac and coagulation function.

Pathophysiology. Heat loss exceeds production; vasoconstriction, ice/crystal and ischemia-reperfusion injure tissue.

Determinants and modifiers. Weather/water, homelessness, intoxication, age, endocrine disease and inadequate protection.

Assessment. Reliable core temperature, ECG, exposure, trauma and local tissue assessment.

Intervention logic. Gentle handling, staged rewarming and specialist tissue/cardiac support by severity.

Known limits and open questions. Temperature devices fail at extremes and apparent death can be misleading.

Suite links. 09 environment; 03A arrhythmia; 03J skin.

Evidence anchors: [16,35]

5.0.9 Hypoglycemia and hyperglycemic crises

Definition and phenotype. Glucose falls below brain needs or rises with insulin deficiency/resistance causing dehydration, acidosis and electrolyte shifts.

Pathophysiology. Medication/fasting/organ failure or insulin deficiency with ketogenesis/hyperosmolality disrupts cellular and whole-body function.

Determinants and modifiers. Diabetes therapy, illness, alcohol, food insecurity, endocrine/organ disease and device error.

Assessment. Immediate glucose plus ketones, electrolytes/osmolality, acid-base, cause and continuous-monitor context.

Intervention logic. Rapid glucose for hypoglycemia; fluids/insulin/electrolytes and cause treatment for crises under protocols.

Known limits and open questions. Interstitial sensors lag and symptoms can occur at different thresholds.

Suite links. 03D endocrine; 03F renal; 03C brain.

Evidence anchors: [12,25]

5.0.10 Orthostatic intolerance and syncope

Definition and phenotype. Symptoms or transient loss of consciousness from impaired cerebral perfusion or reflex/autonomic/cardiac mechanisms on posture or trigger.

Pathophysiology. Volume, venous pooling, baroreflex, heart rate/vascular response and rhythm interact.

Determinants and modifiers. Dehydration, medications, autonomic/neurologic disease, deconditioning, anemia and cardiac disease.

Assessment. Event history/witness, orthostatic vitals, ECG and targeted rhythm/echo/tilt/autonomic testing.

Intervention logic. Treat cause and injury risk; fluid/salt/compression/exercise/medication or cardiac therapy by mechanism.

Known limits and open questions. Wearable HR/HRV alone cannot establish cause; broad syndromic labels mix mechanisms.

Suite links. 03A cardiovascular; 03C nervous; 03G anemia.

Evidence anchors: [2,15,25]

5.0.11 Sleep and circadian disorders

Definition and phenotype. Disrupted sleep timing, continuity, breathing or behavior causes daytime and multisystem impairment.

Pathophysiology. Sleep pressure/clocks, airway, movement, mental health, substances, light and schedule interact.

Determinants and modifiers. Shift work, stress, environment, age, obesity, medications and social constraints.

Assessment. History/diary, actigraphy and polysomnography or other testing by suspected disorder; daytime function.

Intervention logic. Behavioral/circadian interventions, airway/movement treatment and medication selectively.

Known limits and open questions. Consumer stage labels and single-night tests can misrepresent usual sleep.

Suite links. 03B sleep breathing; 05 mental health; 09 shift work.

Evidence anchors: [6,14,17]

5.0.12 Exercise intolerance and exertional symptom syndromes

Definition and phenotype. Reduced ability to sustain activity from cardiopulmonary, muscular, metabolic, autonomic, hematologic, pain or effort/context mechanisms.

Pathophysiology. Oxygen transport/use, ventilatory mechanics, motor capacity, perception and recovery can limit performance.

Determinants and modifiers. Organ disease, deconditioning, anemia, obesity, post-infectious illness, sleep, medications and environment.

Assessment. History, baseline organ assessment, CPET/field or orthostatic testing guided by phenotype and post-exertional recovery.

Intervention logic. Treat mechanism and individualized rehabilitation/pacing; avoid generic exercise prescriptions when post-exertional worsening occurs.

Known limits and open questions. Resting tests often miss limitation and CPET patterns are not always etiologically specific.

Suite links. 03A/B/G/I; 10 rehabilitation; 01 whole-person.

Evidence anchors: [2,7,20]

5.0.13 Fatigue and post-acute recovery disorders

Definition and phenotype. Persistent low energy, effort intolerance or prolonged recovery after illness/injury with heterogeneous mechanisms.

Pathophysiology. Residual organ injury, immune/metabolic/autonomic changes, sleep, mood, pain and deconditioning can coexist.

Determinants and modifiers. Acute severity, reinfection, baseline disease, trauma/stress, social resources and care access.

Assessment. Rule out urgent/treatable disease; longitudinal symptoms/function, sleep, orthostatic/exertional and organ evaluation by phenotype.

Intervention logic. Treat defined disease, conserve/pace or rehabilitate according to response, support sleep/mental/social needs and reassess.

Known limits and open questions. No single biomarker defines fatigue or all post-acute syndromes; treatment evidence is uneven.

Suite links. 01 whole-person; 05/06 psychosocial; 08 infection; 10 rehab.

Evidence anchors: [20,23]

5.0.14 Frailty and delirium vulnerability

Definition and phenotype. Reduced reserve increases disproportionate deterioration under minor stress; delirium is acute fluctuating brain dysfunction.

Pathophysiology. Network deficits across cognition, mobility, nutrition, organs, drugs and environment reduce compensation.

Determinants and modifiers. Age, multimorbidity, sensory loss, hospitalization, sleep disruption, medications and isolation.

Assessment. Baseline function/cognition, frailty and acute change, medication/infection/metabolic evaluation and caregiver history.

Intervention logic. Prevent/resolve precipitants, orient/mobilize, protect sleep/sensory input/nutrition and align care goals.

Known limits and open questions. Scores predict populations but do not replace reversible-cause evaluation or preferences.

Suite links. 07 aging; 03C brain; 12 hospital systems.

Evidence anchors: [20,22,35]

5.0.15 Multiple organ dysfunction and recovery

Definition and phenotype. Progressive failure of several systems during critical illness, followed by variable repair, fibrosis, weakness and cognitive/psychological sequelae.

Pathophysiology. Inflammation, perfusion, iatrogenic exposure, immobilization, sedation, nutrition and organ crosstalk drive trajectory.

Determinants and modifiers. Sepsis/trauma, baseline reserve, treatment timing/intensity and system quality.

Assessment. Serial organ support needs, physiology, exposure timeline, function/cognition and post-discharge outcomes.

Intervention logic. Cause control, protective organ support, early rehabilitation/delirium prevention and longitudinal recovery care.

Known limits and open questions. Organ scores poorly capture lived recovery and comparative critical-care evidence is context dependent.

Suite links. 04 cross-system; 10 rehabilitation; 12 critical care.

Evidence anchors: [18,20,21]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current integrated homeostatic systems risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current integrated homeostatic systems risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[21,23]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[4]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[21]

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Core vital signs and early-warning systems	Noninvasive	HR/ECG, BP, respiratory rate, temperature, SpO2 and mental status.	Basic stability and trend/alert scores.	Ubiquitous and time-critical.	Measurement error, compensatory normality, alarm fatigue and population thresholds.
Fluid balance and body mass	Noninvasive	Recorded intake/output, urine, drains and calibrated weight.	Net balance and trajectory.	Simple longitudinal context.	Recording error, insensible losses and body-composition shifts.
Electrolytes, osmolality and blood gases	Minimally invasive sample	Ion-selective electrodes, osmometers, blood-gas electrochemistry and chemistry.	Na/K/Cl, osmolality, pH, gases, bicarbonate and metabolites.	Directly actionable homeostatic variables.	Sampling error/delay, temperature, heparin/dilution and analyzer variation.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Core and peripheral temperature	Noninvasive to invasive	Thermistors/thermocouples at esophageal, rectal, bladder, tympanic, oral, skin or environmental sites.	Core estimate, gradients and trend.	Essential thermal measurement.	Site lag/bias, sweat, perfusion, environment and extreme-range failure.
Pulse oximetry and blood gas/co-oximetry	Noninvasive/minimally invasive	Minimally length optical pulsatile absorption and direct blood gas/spectrophotometry.	SpO ₂ , PaO ₂ /PaCO ₂ /pH and hemoglobin species.	Continuous trend plus invasive reference.	Motion/perfusion/skin/device bias, dyshe-moglobin and saturation-content confusion.
Hemodynamic ultrasound and invasive monitoring	Noninvasive to invasive	Echo/vascular ultrasound, arterial pressure, central venous/pulmonary artery signals and dilution methods.	Flow, filling/pressure proxies, cardiac function and response.	Mechanism support in shock.	Operator and model assumptions, waveform artifacts, infection/thrombosis.
Continuous glucose and ketone monitoring	Minimally invasive	Electrochemical interstitial glucose and emerging ketone sensing.	Trend, variability, alerts and treatment response.	High temporal resolution.	Lag, compression, calibration/medication interference and alarm burden.
Indirect calorimetry	Noninvasive respiratory measurement	Oxygen consumption and carbon dioxide production from gas flow/concentration.	Energy expenditure and respiratory quotient proxy.	Direct whole-body metabolic rate estimate.	Leaks, non-steady state, ventilator settings and substrate assumptions.
Cardiopulmonary exercise testing	Noninvasive with physiologic stress	Breath-by-breath gas exchange, ECG, BP, oximetry and work rate; sometimes blood gases.	Peak/submaximal capacity, ventilatory, circulatory and metabolic patterns.	Integrated reserve under controlled challenge.	Effort/safety, protocol, calibration and pattern non-specificity.
Orthostatic and autonomic testing	Noninvasive to minimally invasive	Continuous ECG/BP, tilt/stand, breathing/Valsalva, sudomotor and catecholamine tests.	HR/BP reflexes, adrenergic/cardiovagal/sweat responses.	Mechanism support for dysautonomia/syncope.	Protocol, anxiety, hydration, medication and limited ecological validity.
Polysomnography actigraphy and light sensing	Noninvasive	EEG/EOG/EMG, air-flow/effort/SpO ₂ /ECG, movement and ambient light.	Sleep stages, breathing/movement events, timing and rest-activity.	Reference lab plus longitudinal behavior.	Single-night effect, scoring rules, consumer nonequivalence and nonwear.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Wearable multimodal physiology	Noninvasive	PPG/ECG, temperature, EDA, IMU, respiration and environmental channels.	Trends in activity, sleep, stress/recovery and change detection.	Longitudinal context.	Nonspecific inference, device drift, missingness, skin/perfusion bias and behavior effects.
Microcirculation and tissue oxygenation	Noninvasive/capillary	Capillaroscopy, laser Doppler/speckle, NIRS and tissue probes.	Relative flow, oxygenation and capillary morphology.	Potential window beyond macrocirculation.	Depth/scattering, edema, site and uncertain treatment thresholds.
Multiorgan scores and predictive models	Software	Aggregated vitals, labs, support, diagnoses and trajectories.	Severity/risk/alert score.	Summarizes complex data and supports triage/audit.	Circular labels, missingness, treatment effects, bias and automation.

7.0.1 Core vital signs and early-warning systems

Measured quantity and transducer. HR/ECG, BP, respiratory rate, temperature, SpO2 and mental status.

Typical outputs. Basic stability and trend/alert scores.

Strengths. Ubiquitous and time-critical.

Limitations and artifacts. Measurement error, compensatory normality, alarm fatigue and population thresholds.

Reference and validation strategy. Metrology, manual confirmation, workflow and outcome trials.

Clinical/research maturity. Established.

Evidence anchors: [29,35]

7.0.2 Fluid balance and body mass

Measured quantity and transducer. Recorded intake/output, urine, drains and calibrated weight.

Typical outputs. Net balance and trajectory.

Strengths. Simple longitudinal context.

Limitations and artifacts. Recording error, insensible losses and body-composition shifts.

Reference and validation strategy. Scale calibration, reconciliation and clinical response.

Clinical/research maturity. Established but error-prone.

Evidence anchors: [10,35]

7.0.3 Electrolytes, osmolality and blood gases

Measured quantity and transducer. Ion-selective electrodes, osmometers, blood-gas electrochemistry and chemistry.

Typical outputs. Na/K/Cl, osmolality, pH, gases, bicarbonate and metabolites.

Strengths. Directly actionable homeostatic variables.

Limitations and artifacts. Sampling air/delay, temperature, heparin/dilution and analyzer variation.

Reference and validation strategy. Traceable controls, preanalytics and method comparison.

Clinical/research maturity. Established.

Evidence anchors: [3,27]

7.0.4 Core and peripheral temperature

Measured quantity and transducer. Thermistors/thermocouples/infrared at esophageal, rectal, bladder, tympanic, oral, skin or environmental sites.

Typical outputs. Core estimate, gradients and trend.

Strengths. Essential thermal measurement.

Limitations and artifacts. Site lag/bias, sweat, perfusion, environment and extreme-range failure.

Reference and validation strategy. Blackbody/water-bath calibration and clinical site comparison across conditions.

Clinical/research maturity. Established with site limits.

Evidence anchors: [16,25]

7.0.5 Pulse oximetry and blood gas/co-oximetry

Measured quantity and transducer. Multiwavelength optical pulsatile absorption and direct blood gas/spectrophotometry.

Typical outputs. SpO₂, PaO₂/PaCO₂/pH and hemoglobin species.

Strengths. Continuous trend plus invasive reference.

Limitations and artifacts. Motion/perfusion/skin/device bias, dyshemoglobin and saturation-content confusion.

Reference and validation strategy. Bench simulators plus paired arterial co-oximetry across sub-groups/conditions.

Clinical/research maturity. Established.

Evidence anchors: [7,25,34]

7.0.6 Hemodynamic ultrasound and invasive monitoring

Measured quantity and transducer. Echo/vascular ultrasound, arterial pressure, central venous/pulmonary artery signals and dilution methods.

Typical outputs. Flow, filling/pressure proxies, cardiac function and response.

Strengths. Mechanism support in shock.

Limitations and artifacts. Operator and model assumptions, waveform artifacts, infection/thrombosis.

Reference and validation strategy. Phantoms, waveform/calibration, expert comparison and outcome-linked protocols.

Clinical/research maturity. Established selected.

Evidence anchors: [2,29]

7.0.7 Continuous glucose and ketone monitoring

Measured quantity and transducer. Electrochemical interstitial glucose and emerging ketone sensing.

Typical outputs. Trend, variability, alerts and treatment response.

Strengths. High temporal resolution.

Limitations and artifacts. Lag, compression, calibration/medication interference and alarm burden.

Reference and validation strategy. Bench interference, paired reference and clinical utility.

Clinical/research maturity. Established glucose; ketones emerging.

Evidence anchors: [12,25]

7.0.8 Indirect calorimetry

Measured quantity and transducer. Oxygen consumption and carbon dioxide production from gas flow/concentration.

Typical outputs. Energy expenditure and respiratory quotient proxy.

Strengths. Direct whole-body metabolic rate estimate.

Limitations and artifacts. Leaks, non-steady state, ventilator settings and substrate assumptions.

Reference and validation strategy. Gas/flow calibration and controlled protocol.

Clinical/research maturity. Established specialist/research.

Evidence anchors: ^[3,7]

7.0.9 Cardiopulmonary exercise testing

Measured quantity and transducer. Breath-by-breath gas exchange, ECG, BP, oximetry and work rate; sometimes blood gases.

Typical outputs. Peak/submaximal capacity, ventilatory, circulatory and metabolic patterns.

Strengths. Integrated reserve under controlled challenge.

Limitations and artifacts. Effort/safety, protocol, calibration and pattern non-specificity.

Reference and validation strategy. Daily gas/flow calibration, repeatability and clinical outcome correlation.

Clinical/research maturity. Established specialist.

Evidence anchors: ^[2,7]

7.0.10 Orthostatic and autonomic testing

Measured quantity and transducer. Continuous ECG/BP, tilt/stand, breathing/Valsalva, sudomotor and catecholamine tests.

Typical outputs. HR/BP reflexes, adrenergic/cardiovagal/sweat responses.

Strengths. Mechanism support for dysautonomia/syncope.

Limitations and artifacts. Protocol, anxiety, hydration, medication and limited ecological validity.

Reference and validation strategy. Signal calibration, normative protocol and symptom/outcome linkage.

Clinical/research maturity. Established specialist.

Evidence anchors: [2,15]

7.0.11 Polysomnography, actigraphy and light sensing

Measured quantity and transducer. EEG/EOG/EMG, airflow/effort/SpO2/ECG, movement and ambient light.

Typical outputs. Sleep stages, breathing/movement events, timing and rest-activity.

Strengths. Reference lab plus longitudinal behavior.

Limitations and artifacts. Single-night effect, scoring rules, consumer nonequivalence and nonwear.

Reference and validation strategy. AASM standards, clock synchronization and outcome utility.

Clinical/research maturity. Established; consumer staging limited.

Evidence anchors: [14,25]

7.0.12 Wearable multimodal physiology

Measured quantity and transducer. PPG/ECG, temperature, EDA, IMU, respiration and environmental channels.

Typical outputs. Trends in activity, sleep, stress/recovery and change detection.

Strengths. Longitudinal context.

Limitations and artifacts. Nonspecific inference, device drift, missingness, skin/perfusion bias and behavior effects.

Reference and validation strategy. Raw data/quality, reference challenges, subgroup and clinical utility.

Clinical/research maturity. Emerging.

Evidence anchors: [25,32]

7.0.13 Microcirculation and tissue oxygenation

Measured quantity and transducer. Capillaroscopy, laser Doppler/speckle, NIRS and tissue probes.

Typical outputs. Relative flow, oxygenation and capillary morphology.

Strengths. Potential window beyond macrocirculation.

Limitations and artifacts. Depth/scattering, edema, site and uncertain treatment thresholds.

Reference and validation strategy. Optical/flow phantoms and outcome-linked prospective validation.

Clinical/research maturity. Research/selected clinical.

Evidence anchors: [2,27]

7.0.14 Multiorgan scores and predictive models

Measured quantity and transducer. Aggregated vitals, labs, support, diagnoses and trajectories.

Typical outputs. Severity/risk/alert score.

Strengths. Summarizes complex data and supports triage/audit.

Limitations and artifacts. Circular labels, missingness, treatment effects, bias and automation.

Reference and validation strategy. Temporal external validation, calibration, subgroup and decision-impact trials.

Clinical/research maturity. Established scores; AI emerging.

Evidence anchors: [32,33]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise integrated homeostatic systems quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise integrated homeostatic systems quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [25,28]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [28,29]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[27,30]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[27]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [32,33]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [32]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [26,29,31]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Remove cause and restore first principles	Acute treatment	Control bleeding/infection/obstruction and restore oxygen, flow, temperature, glucose/electrolytes as needed.	Causal source control/exposure dominates supportive signal normalization.	Delay, treatment conflict and overcorrection.	Cause control, physiology, organ function and adverse effects.
Fluid, electrolyte and acid-base therapy	Acute/chronic	Replace or remove water/solute and correct dangerous disturbances at safe rates.	Life-saving when mechanism and rate are correct.	Overload, osmotic demyelination/cerebral edema, arrhythmia and iatrogenic shifts.	Serial labs, weight/output, neurologic/cardiopulmonary status and cause.
Oxygen and ventilatory support	Acute/chronic	Increase inspired oxygen or support ventilation/airway.	Essential for defined hypoxemia/ventilatory failure.	Hyperoxia, pressure/volume injury, CO2 retention, device dependence and delayed cause treatment.	Oxygenation, ventilation, work, organ response and device harms.
Hemodynamic support	Acute	Use positioning, fluid/blood, vasoactive/inotropic and mechanical support according to shock mechanism.	Can restore perfusion while source control proceeds.	Arrhythmia, ischemia, overload, limb/device complications and false target chasing.	Perfusion, organ response, flow/pressure and adverse effects.
Temperature management	Acute/prevention	Rapid cooling or controlled rewarming plus environmental/work changes.	Time-critical in heat stroke/hypothermia and preventable through systems.	Cold/rewarming injury and inadequate field measurement.	Core temperature, neurologic/organ status and exposure recurrence.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Sleep/circadian and recovery intervention	Prevention/treatment	Align timing/light/behavior, treat sleep disorders and redesign shift-work or care environments.	Improves defined sleep outcomes and supports broader health.	Sedative harm, schedule infeasibility and worker inequity.	Sleep timing/quality, daytime function, safety and adherence.
Exercise, pacing and rehabilitation	Prevention/rehabilitation	Build resilience and function or manage energy within physiologic tolerance.	Strong across many chronic conditions; dose and phenotype matter.	Falls, overexertion, post-exertional worsening and inaccessible programs.	Symptoms, recovery, capacity, activity and participation.
Nutrition and metabolic support	Treatment/prevention	Provide adequate energy/protein/micronutrients while avoiding overfeeding and respecting disease/feeding capacity.	Improves defined deficiency and recovery outcomes; critical-care specifics vary.	Refeeding, aspiration, glycemic/electrolyte effects and formula/food inequity.	Intake, weight/body composition, electrolytes, function and tolerance.
Delirium, mobility and iatrogenesis prevention	System/rehabilitation	Reduce unnecessary sedation/restraints/devices, orient, mobilize and preserve sensory/sleep input.	Bundles reduce selected hospital harms.	Staffing burden and protocol inconsistency.	Delirium, mobility, falls, device days, sleep and function.
Multidisciplinary critical-care and recovery pathways	System	Coordinate organ support, rehabilitation, mental health, family, primary care and palliative goals.	Addresses survival plus post-intensive-care outcomes.	Fragmentation and resource inequity.	Survival, readmission, cognition, mental health, function and participation.
Environmental and occupational control	Upstream prevention	Modify heat/cold, air, work-rest, hydration, shift schedule, PPE and built environment.	Prevents many homeostatic failures more effectively than individual monitoring alone.	Employer/housing power imbalance and implementation gaps.	Exposure, illness/injury, worker outcomes and equity.
Closed-loop and decision-support systems	Advanced intervention	Automatically adjust insulin, ventilation, temperature or other support from sensors/models.	Established for narrow systems; potential to reduce workload and variability.	Sensor/model failure, runaway control, cybersecurity, mode confusion and automation bias.	Bench fault injection, human factors, supervised trials and postmarket incidents.

9.0.1 Remove cause and restore first principles

Target and mechanism. Control bleeding/infection/obstruction/toxin/exposure and restore oxygen, flow, temperature, glucose/electrolytes as needed.

Evidence and expected outcome. Causal source control usually dominates supportive signal normalization.

Risks, burden, and equity. Delay, treatment conflict and overcorrection.

Response monitoring. Cause control, physiology, organ function and adverse effects.

Evidence anchors: [18,35]

9.0.2 Fluid, electrolyte and acid-base therapy

Target and mechanism. Replace or remove water/solute and correct dangerous disturbances at safe rates.

Evidence and expected outcome. Life-saving when mechanism and rate are correct.

Risks, burden, and equity. Overload, osmotic demyelination/cerebral edema, arrhythmia and iatrogenic shifts.

Response monitoring. Serial labs, weight/output, neurologic/cardiopulmonary status and cause.

Evidence anchors: [3,10]

9.0.3 Oxygen and ventilatory support

Target and mechanism. Increase inspired oxygen or support ventilation/airway.

Evidence and expected outcome. Essential for defined hypoxemia/ventilatory failure.

Risks, burden, and equity. Hyperoxia, pressure/volume injury, CO₂ retention, device dependence and delayed cause treatment.

Response monitoring. Oxygenation, ventilation, work, organ response and device harms.

Evidence anchors: [7,8]

9.0.4 Hemodynamic support

Target and mechanism. Use positioning, fluid/blood, vasoactive/inotropic and mechanical support according to shock mechanism.

Evidence and expected outcome. Can restore perfusion while source control proceeds.

Risks, burden, and equity. Arrhythmia, ischemia, overload, limb/device complications and false target chasing.

Response monitoring. Perfusion, organ response, flow/pressure and adverse effects.

Evidence anchors: [2,18]

9.0.5 Temperature management

Target and mechanism. Rapid cooling or controlled rewarming plus environmental/work changes.

Evidence and expected outcome. Time-critical in heat stroke/hypothermia and preventable through systems.

Risks, burden, and equity. Cold/rewarming injury and inadequate field measurement.

Response monitoring. Core temperature, neurologic/organ status and exposure recurrence.

Evidence anchors: [16,17]

9.0.6 Sleep/circadian and recovery intervention

Target and mechanism. Align timing/light/behavior, treat sleep disorders and redesign shift-work or care environments.

Evidence and expected outcome. Improves defined sleep outcomes and supports broader health.

Risks, burden, and equity. Sedative harm, schedule infeasibility and worker inequity.

Response monitoring. Sleep timing/quality, daytime function, safety and adherence.

Evidence anchors: [14,17]

9.0.7 Exercise, pacing and rehabilitation

Target and mechanism. Build reserve and function or manage energy within physiologic tolerance.

Evidence and expected outcome. Strong across many chronic conditions; dose and phenotype matter.

Risks, burden, and equity. Falls, overexertion, post-exertional worsening and inaccessible programs.

Response monitoring. Symptoms, recovery, capacity, activity and participation.

Evidence anchors: [20,21]

9.0.8 Nutrition and metabolic support

Target and mechanism. Provide adequate energy/protein/micronutrients while avoiding overfeeding and respecting disease/feeding capacity.

Evidence and expected outcome. Improves defined deficiency and recovery outcomes; critical-care specifics vary.

Risks, burden, and equity. Refeeding, aspiration, glycemic/electrolyte effects and formula/food inequity.

Response monitoring. Intake, weight/body composition, electrolytes, function and tolerance.

Evidence anchors: [13]

9.0.9 Delirium, mobility and iatrogenesis prevention

Target and mechanism. Reduce unnecessary sedation/restraints/devices, orient, mobilize and preserve sensory/sleep input.

Evidence and expected outcome. Bundles reduce selected hospital harms.

Risks, burden, and equity. Staffing burden and protocol inconsistency.

Response monitoring. Delirium, mobility, falls, device days, sleep and function.

Evidence anchors: [20,35]

9.0.10 Multidisciplinary critical-care and recovery pathways

Target and mechanism. Coordinate organ support, rehabilitation, mental health, family, primary care and palliative goals.

Evidence and expected outcome. Addresses survival plus post-intensive-care outcomes.

Risks, burden, and equity. Fragmentation and resource inequity.

Response monitoring. Survival, readmission, cognition, mental health, function and participation.

Evidence anchors: [20,24,36]

9.0.11 Environmental and occupational control

Target and mechanism. Modify heat/cold, air, work-rest, hydration, shift schedule, PPE and built environment.

Evidence and expected outcome. Prevents many homeostatic failures more effectively than individual monitoring alone.

Risks, burden, and equity. Employer/housing power imbalance and implementation gaps.

Response monitoring. Exposure, illness/injury, worker outcomes and equity.

Evidence anchors: [16,17]

9.0.12 Closed-loop and decision-support systems

Target and mechanism. Automatically adjust insulin, ventilation, temperature or other support from sensors/models.

Evidence and expected outcome. Established for narrow systems; potential to reduce workload and variability.

Risks, burden, and equity. Sensor/model failure, runaway control, cybersecurity, mode confusion and automation bias.

Response monitoring. Bench fault injection, human factors, supervised trials and postmarket incidents.

Evidence anchors: [26,28,29,30]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can multimodal wearables detect decompensation before symptoms?	Continuous signals reveal within-person change.	Reliable actionable transitions across illnesses and contexts.	Nonspecific changes, missingness and response pathways drive false alerts.	Pragmatic randomized monitoring trials with net benefit and subgroup analysis.
Can a physiological digital twin guide therapy?	Mechanistic models can fit selected organ systems and controlled challenges.	Individual identifiability and benefit under changing disease/treatment.	Many latent parameters produce similar outputs and models drift.	Prospective falsification across held-out perturbations before decision-impact trials.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within integrated homeostatic systems.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can multimodal wearables detect decompensation before symptoms?

Established knowledge. Continuous signals reveal within-person change.

Unresolved issue. Reliable actionable transitions across illnesses and contexts.

Why current evidence is insufficient. Nonspecific changes, missingness and response pathways drive false alerts.

Research that would change confidence. Pragmatic randomized monitoring trials with net benefit and subgroup analysis.

Evidence anchors: [25,32]

10.0.2 Can a physiological digital twin guide therapy?

Established knowledge. Mechanistic models can fit selected organ systems and controlled challenges.

Unresolved issue. Individual identifiability and benefit under changing disease/treatment.

Why current evidence is insufficient. Many latent parameters produce similar outputs and models drift.

Research that would change confidence. Prospective falsification across held-out perturbations before decision-impact trials.

Evidence anchors: ^[5]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within integrated homeostatic systems.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[23,32]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: [25,33]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: [21,23]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Open multimodal challenge dataset	What signatures distinguish limited flow, ventilation, muscle, autonomic and metabolic reserve?	Synchronized CPET/orthostatic/thermal or meal challenges with reference signals, symptoms, recovery and diverse participants.	Reproducible interpretable mechanisms and external validation.	Black-box clusters without causal or treatment relevance.
0. Closed-loop safety benchmark	Can fault-tolerant control abstain safely under sensor failure and patient-state shift?	Hardware-in-loop simulators, formal hazards, fault injection, human factors and staged clinical trials.	No unsafe actuation under predefined faults and improved out-comes/workload.	Silent coupling of sensor/model error to harmful therapy.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for integrated homeostatic systems?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Open multimodal challenge dataset

Question. What signatures distinguish limited flow, ventilation, muscle, autonomic and metabolic reserve?

Design. Synchronized CPET/orthostatic/thermal or meal challenges with reference signals, symptoms, recovery and diverse participants.

Success criterion. Reproducible interpretable mechanisms and external validation.

Failure or stop criterion. Black-box clusters without causal or treatment relevance.

Evidence anchors: [2,7]

11.0.2 0. Closed-loop safety benchmark

Question. Can fault-tolerant control abstain safely under sensor failure and patient-state shift?

Design. Hardware-in-loop simulators, formal hazards, fault injection, human factors and staged clinical trials.

Success criterion. No unsafe actuation under predefined faults and improved outcomes/workload.

Failure or stop criterion. Silent coupling of sensor/model error to harmful therapy.

Evidence anchors: [26,28,30]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for integrated homeostatic systems?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: ^[34]

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[28,32]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[36]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: ^[21]

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[34,36]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Flow, pressure, oxygen delivery, autonomic control and shock.	hemodynamics or perfusion is central.
03B	Respiratory Health	Ventilation, gas exchange, acid-base, sleep and exercise.	oxygen/CO2 or ventilatory reserve is central.
03F	Renal and Urinary Health	Fluid, electrolytes, acid-base and toxin clearance.	volume or chemistry is changing.
04	Cross-System Pathological Processes	Sepsis, inflammation, fibrosis, shock and multiorgan failure.	homeostatic failure becomes systemic pathology.
10	Function, Disability and Rehabilitation	Reserve, fatigue, frailty and recovery require person-centered functional outcomes.	physiology must translate into capability.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Homeostasis	Dynamic regulation of variables within viable ranges.
Allostasis	Adaptive change in regulatory set and effort to meet anticipated or actual demand.
Reserve	Capacity to increase or sustain function under challenge beyond resting baseline.
Shock	Circulatory failure causing inadequate or maldistributed tissue perfusion and oxygen/substrate delivery.
Resilience	Ability of a system or person to resist, recover from or adapt to stress while preserving valued function.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.